Article

AERIS: Environment-Aware Hierarchical Routing for Reliable Wireless Sensor Networks under Realistic Channels

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Abstract: This paper evaluates AERIS, a lightweight hierarchical routing protocol for wireless sensor networks under heterogeneous channel conditions. The primary metric is `pdr_expected` (base-station delivered packets divided by expected source packets). In the publication-tier 100-node matrix (4 environments, 30 seeds per environment), AERIS achieves the highest packet delivery ratio within the tested baseline set (LEACH, PEGASIS, HEED, TEEN). In the large-scale matrix (100–1000 nodes; indoor_factory n=650, other environments n=550), AERIS ranks first in three environments (indoor_factory, outdoor_urban, outdoor_suburban), while PEGASIS remains higher in indoor_office. Ablation indicates environment-dependent module contribution: Gateway is positive in three environments and near-neutral in indoor_office; CAS does not provide a consistent positive marginal effect. Hop-based latency analysis shows AERIS near two hops, while PEGASIS remains above 31 hops due to chain traversal. NS-3 evidence is used strictly as trend-level cross-platform validation (AERIS vs LEACH), not as numerical-equivalence proof. The resulting claim is intentionally scoped: AERIS is effective for reliability-oriented deployments in heterogeneous channels, but not a universal best protocol across all environment-scale combinations.

Keywords: wireless sensor networks; reliable routing; hierarchical protocol; packet delivery ratio; scalability; reproducible evaluation

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1. Introduction

Wireless sensor network (WSN) routing is classically optimized for energy reduction, but practical deployments also require stable delivery under heterogeneous channels [1,2]. This creates a multi-objective tension among reliability, hop-based latency, and energy use. Existing protocols represent distinct operating points: LEACH emphasizes clustered forwarding [3], PEGASIS emphasizes chain-based aggregation [4], HEED emphasizes residual-energy-aware head selection [5], and TEEN emphasizes event-triggered reporting [6].

AERIS is designed as a lightweight hierarchical protocol that prioritizes reliability under realistic channel variability. The design objective is not universal dominance, but robust performance across multiple channel environments while maintaining deployability on resource-constrained nodes.

Contributions and Scope Boundaries

This manuscript makes three scoped contributions. First, it provides publication-tier evidence for multi-environment reliability comparison at 100 nodes and environment-stratified scalability analysis up to 1000 nodes. Second, it quantifies module-level behavior with ablation under the same channel taxonomy. Third, it adds cross-platform trend validation in NS-3 for AERIS versus LEACH.

The manuscript explicitly does not claim universal protocol superiority or cross-platform numerical equivalence.

2. Related Work

Classical WSN protocols (LEACH, PEGASIS, HEED, TEEN) provide foundational routing mechanisms but show different failure modes under harsh channels and larger scales [4,5,7]. Chain-based aggregation can reduce per-hop transmission effort but increases end-to-end hop count. Cluster-based approaches can reduce hop count but may become sensitive to uplink quality and head placement.

Recent adaptive or learning-based methods improve specific settings but often require heavier runtime or training overhead [8,9]. In this manuscript, the focus is a rule-based protocol with explicit reproducibility constraints and auditable experiment metadata.

3. System Model and Protocol

3.1. Metric Definition

The primary metric is

$$\mathrm{PDR}_{\mathrm{expected}} = \frac{\mathrm{bs\_delivered}}{\mathrm{source\_packets\_expected}}. \tag{1}$$

All publication conclusions in this draft are based on this metric.

3.2. AERIS Architecture

AERIS combines three components:

1. Context-Adaptive Switching (CAS) for mode control,
2. Skeleton backbone for hierarchical forwarding,
3. Gateway coordination for CH-to-BS reinforcement.

Gateway candidate scoring follows a weighted rule

$$\mathrm{Score}_i = \alpha E_i + \beta C_i + \gamma L_i, \tag{2}$$

where E_i is residual energy, C_i is centrality, and L_i is link quality.

4. Experimental Setup

4.1. Publication-Tier Settings

- Simulator: Python implementation with logged provenance.
- Channel environments: indoor_office, indoor_factory, outdoor_urban, outdoor_suburban.
- Default node count: 100.
- Seeds for multi-environment and ablation: 30 independent seeds.
- Seeds for scalability: environment-specific sample sizes with n=550 (indoor_factory uses n=650).
- Baselines: LEACH, PEGASIS, HEED, TEEN.

4.2. Evidence Scope

The analysis is grounded in publication-tier multi-environment comparison, ablation, scalability, hop-based latency, energy-lifetime statistics, and NS-3 trend validation. Claims are constrained to this evidence scope. Cross-platform numerical equivalence is intentionally not claimed.

4.3. Statistical Methods

For pairwise protocol comparisons, we use Welch's two-sample t-test because variance equality is not assumed across protocols and environments. Family-wise control is handled by Holm correction over each comparison family. Effect size is reported with Hedges' g and interpreted as: small (0.2), medium (0.5), and large (0.8) in absolute value. Confidence intervals are reported as two-sided 95% intervals. All significance statements in this manuscript are based on Holm-corrected p values.

4.4. Reproducibility Controls

All publication-tier scripts in this study use deterministic seed lists. They also enforce `force_ctp_reliable=False`. This ensures that PDR reflects simulated delivery behavior rather than any reliability override. We report `pdr_expected` as the primary metric and treat hop count as a latency proxy, not a wall-clock latency measurement.

5. Results

5.1. Multi-Environment Comparison at 100 Nodes (n=30)

Table 1. PDR by environment at 100 nodes (mean ± std; std reported as population standard deviation, ddof=0).

Environment	AERIS	LEACH	PEGASIS	HEED	TEEN
indoor_office	0.9739±0.0047	0.5543±0.0401	0.9078±0.0166	0.9371±0.0076	0.8222±0.0044
indoor_factory	0.6031±0.0258	0.1614±0.0209	0.1928±0.0255	0.2326±0.0263	0.3113±0.0245
outdoor_urban	0.3745±0.0354	0.0552±0.0127	0.0542±0.0117	0.0635±0.0121	0.1201±0.0183
outdoor_suburban	0.7451±0.0193	0.2703±0.0272	0.3382±0.0329	0.4221±0.0313	0.4752±0.0236

At 100 nodes, AERIS ranks first in all four tested environments within the evaluated baseline set (LEACH, PEGASIS, HEED, TEEN). This claim is restricted to the 100-node matrix and should not be generalized to the 1000-node regime.

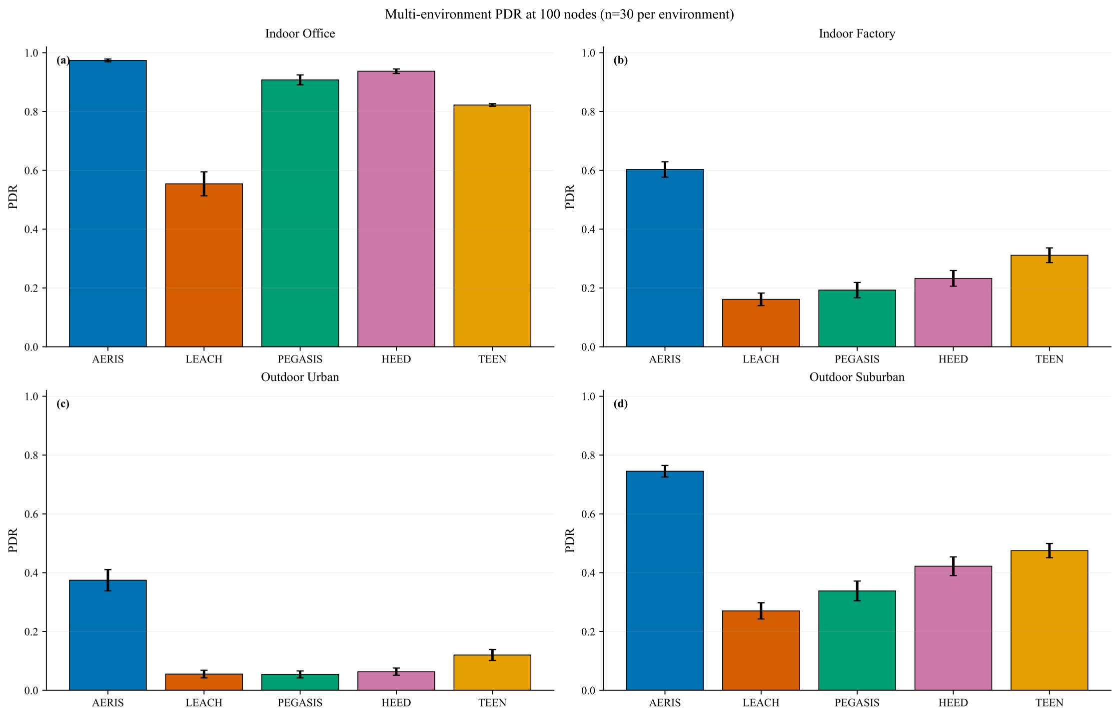

Figure 1. PDR comparison panel at 100 nodes (4 environments, n=30).

5.2. Ablation: Environment-Dependent Effects (n=30)

Table 2. Full vs no_gateway PDR (mean ± std; std reported as population standard deviation, ddof=0).

Environment	Full	no_gateway	Delta (no_gateway - full)
indoor_office	0.9739±0.0047	0.9741±0.0036	+0.0002
indoor_factory	0.6031±0.0258	0.5806±0.0215	-0.0225
outdoor_urban	0.3745±0.0354	0.3534±0.0301	-0.0212
outdoor_suburban	0.7451±0.0193	0.7306±0.0264	-0.0146

Gateway is positive in three environments and near-neutral in indoor_office. CAS is not consistently positive across environments. In indoor_office, the Full-vs-no_gateway gap is +0.0002 (absolute PDR), so this cell is treated as practically neutral.

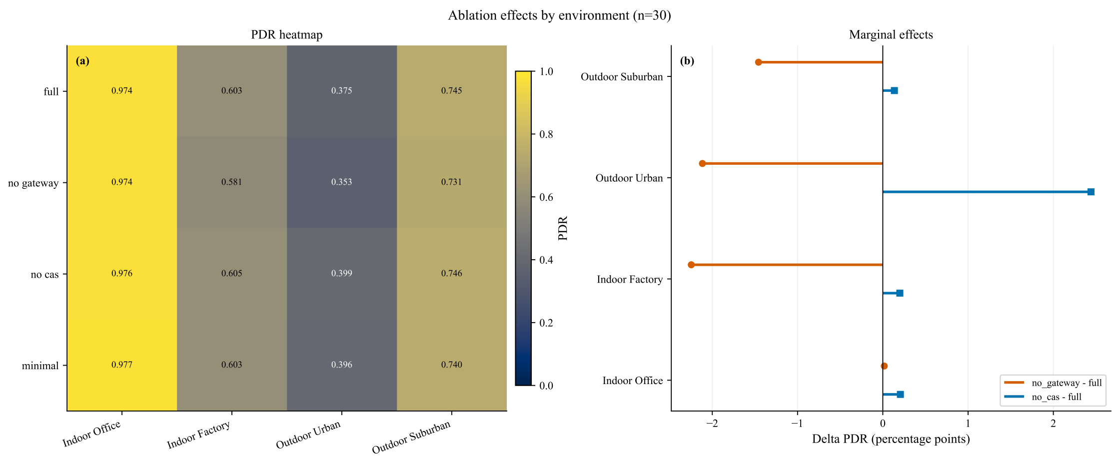

Figure 2. Ablation panel: heatmap and marginal effects (n=30).

5.3. Scalability (100–1000 Nodes; n=550/650)

Table 3. PDR ranking at 1000 nodes (mixed sample sizes: n=650 in indoor_factory, n=550 in other environments).

Environment	AERIS	LEACH	PEGASIS	HEED	TEEN	AERIS rank
indoor_office	0.9899	0.9902	0.9991	0.9911	0.9922	5th
indoor_factory	0.9725	0.1888	0.1619	0.1631	0.2196	1st
outdoor_urban	0.8849	0.0623	0.0487	0.0341	0.0664	1st
outdoor_suburban	0.9900	0.5952	0.7866	0.5544	0.6899	1st

AERIS is first in 3/4 environments in the large-scale matrix, while PEGASIS is higher in indoor_office. This statement is matrix-specific and should not be interpreted as a pooled cross-environment ranking. Because sample sizes differ by environment (n=550 or n=650), all large-scale claims are reported per environment and are not pooled into a single global estimate.

5.4. Statistical Robustness Snapshot (AERIS vs LEACH)

Table 4. Representative Holm-corrected significance snapshot from scalability matrix.

Environment	Nodes	ΔPDR	Hedges' g	Holm p	Significant
indoor_office	100	+0.004548	+2.3398	1.32×10^{-207}	yes
indoor_office	500	-0.000130	-0.1009	9.46×10^{-2}	no
indoor_office	1000	-0.000286	-0.2673	1.05×10^{-5}	yes
indoor_factory	1000	+0.783773	+151.3826	$< 10^{-300}$	yes
outdoor_urban	1000	+0.822585	+181.4951	$< 10^{-300}$	yes
outdoor_suburban	1000	+0.394856	+65.3769	$< 10^{-300}$	yes

This snapshot highlights the scoped conclusion used throughout this manuscript: strong and consistent gains in harsh environments, but environment-dependent behavior in benign indoor_office conditions. For very large sample sizes, tiny mean differences can still produce small p values; therefore, practical interpretation in this study follows both effect size and absolute PDR gap, not significance labels alone (for example, indoor_office at 1000 nodes has a small negative delta versus LEACH). The very large absolute values of Hedges' g in harsh environments reflect both large mean gaps and low within-group variance under large sample sizes; they should be interpreted as magnitude indicators within this experiment design, not as cross-study constants.

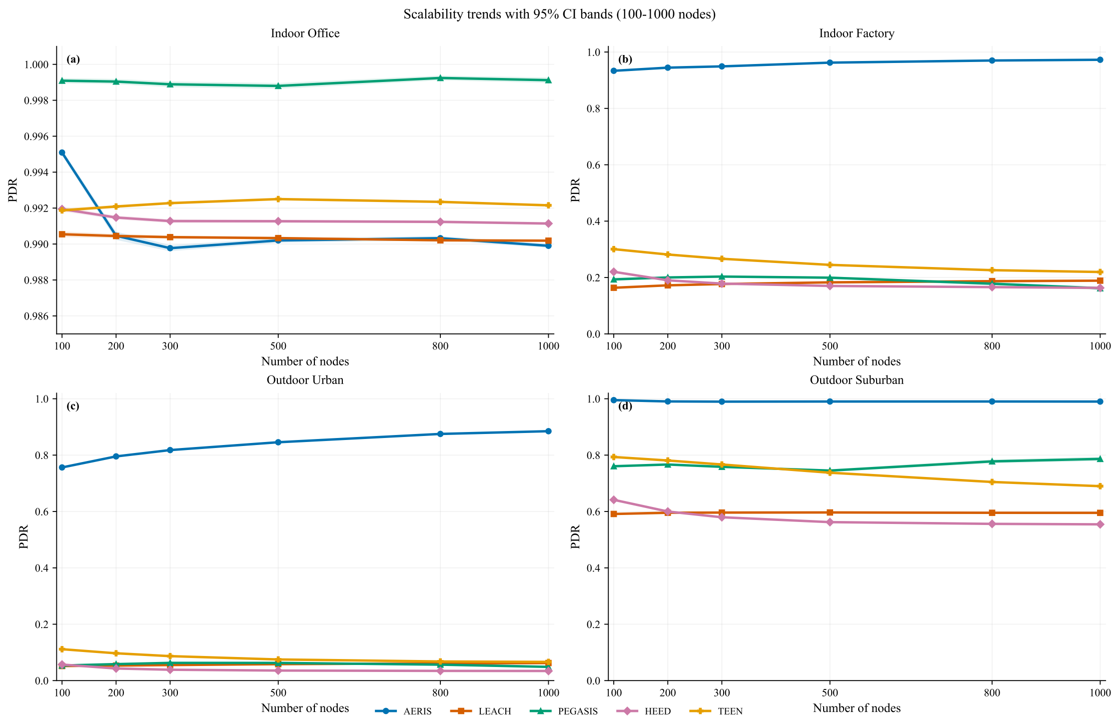

Figure 3. Scalability trends over node counts (indoor_factory n=650, other environments n=550) with 95% CI bands. The indoor_office panel uses a narrowed y-axis window to expose small but non-zero ordering gaps near PDR=1.0.

5.5. Hop-Based Latency and Trade-offs

At 100 nodes (n=30), AERIS stays near two hops (1.97–1.99), while PEGASIS remains above 31 hops due to chain traversal. LEACH and TEEN show lower average hops because of direct-to-BS transmissions in their protocol behavior.

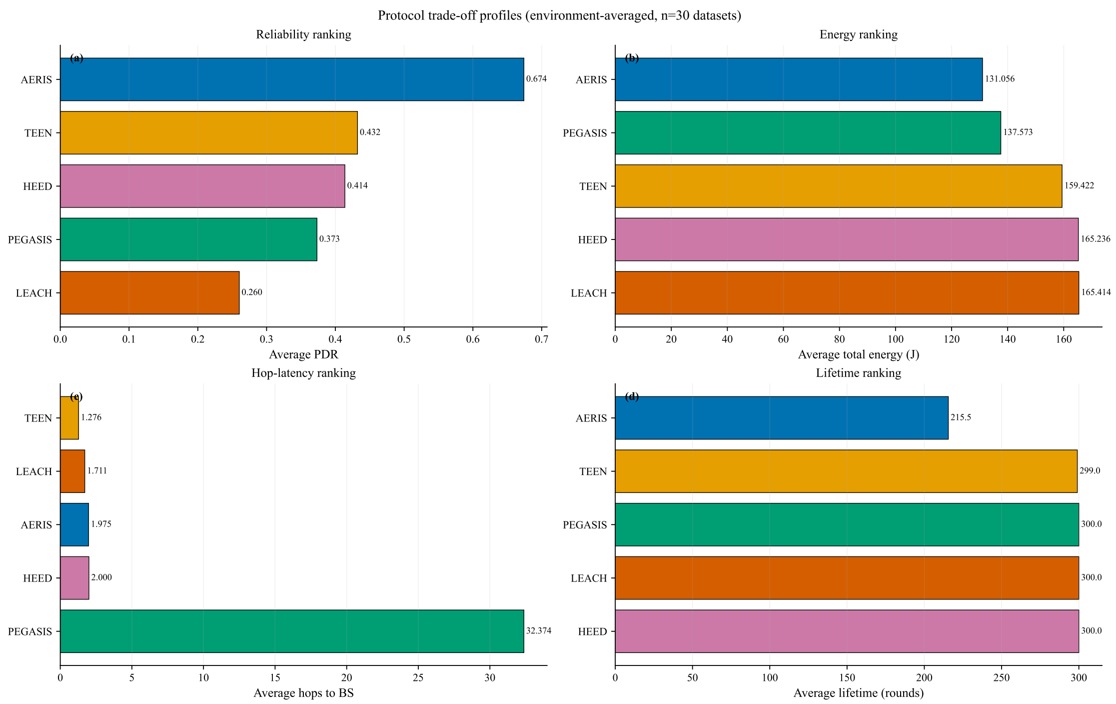

Figure 4. Trade-off panel from publication-tier data: reliability, energy, hop-based latency, and lifetime.

5.6. NS-3 Trend Validation (AERIS vs LEACH, n=30)

NS-3 validation is used as a cross-platform trend check rather than a numerical-equivalence proof. This block is restricted to AERIS-versus-LEACH. Under aligned multi-environment settings, AERIS is directionally not lower than LEACH in all tested environment-scale cells. At 100 nodes, significance is confirmed in 3/4 environments (indoor_factory, outdoor_urban, outdoor_suburban), while indoor_office remains non-significant. In the 300/500-node extension, all 8 environment-scale comparisons are significant after Holm correction.

Table 5. NS-3 trend-level validation at 100 nodes (AERIS vs LEACH, n=30; metric=`pdr_expected`).

Environment	AERIS	LEACH	Delta	Holm-corrected p	Significance
indoor_office	0.9202	0.9185	+0.0017	1.00	no
indoor_factory	0.5991	0.5330	+0.0661	$< 10^{-20}$	yes
outdoor_urban	0.2057	0.1889	+0.0169	7.89×10^{-4}	yes
outdoor_suburban	0.7767	0.6929	+0.0838	3.17×10^{-25}	yes

Scope note: this NS-3 block supports trend consistency only. The manuscript does not claim cross-platform numerical equivalence. Across 5 node counts and 4 environments, 18/20 AERIS-versus-LEACH comparisons are significant.

6. Discussion

The evidence supports a scoped positioning rather than universal superiority. AERIS is strong for reliability across heterogeneous channels, but it is not the top protocol in every environment-scale pair. Indoor_office is a stable counterexample where PEGASIS is higher in the scalability matrix.

Practical interpretation requires separating statistical and engineering significance. At very large sample sizes, tiny absolute gaps can be statistically significant; therefore, conclusions in this paper rely on both corrected p values and absolute PDR deltas. This is particularly relevant for indoor_office at large scale, where ordering differences are small in magnitude.

Energy interpretation also requires caution: lower total energy can co-occur with shorter lifetime if a protocol terminates earlier. For this reason, reliability-energy statements are framed as conditioned trade-offs under fixed evaluation settings, not universal dominance claims.

The manuscript intentionally keeps three evidence blocks separate:

- 100-node multi-environment comparison,
- large-scale matrix (100–1000 nodes) with mixed environment-specific sample sizes,
- NS-3 trend validation for AERIS versus LEACH only.

Cross-block numerical pooling is avoided.

7. Limitations and Future Work

- Current evidence is simulator-based.
- NS-3 trend-level publication gate is completed; numerical equivalence is intentionally not claimed because of platform-depth differences.
- The large-scale matrix uses environment-specific sample sizes (n=550 or n=650), so conclusions are not presented as a pooled single-regime estimate.
- Reported gains are bounded to the tested baseline set and channel taxonomy.
- Additional topology families (corridor, hotspot) and hardware-in-the-loop validation remain pending.

8. Conclusion

AERIS provides a reliability-oriented routing option for heterogeneous WSN channels. In the publication-tier 100-node matrix, it achieves the highest PDR within the tested baseline set across all four environments. In the large-scale matrix, it remains first in three of four environments, with indoor_office as a stable counterexample where PEGASIS is higher. NS-3 confirms trend consistency for AERIS versus LEACH and is used strictly as trend-level evidence. Gateway effects are environment-dependent and CAS effects are mixed; therefore, the final claim is explicitly bounded to the tested protocol set, environments, metrics, and sample sizes.

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Data Availability Statement

The datasets, scripts, and manuscript sources supporting this study are available from the corresponding author and will be released in the public project repository upon final publication.

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